

Problem Z. Fibonacci Numbers

Time Limit 1000 ms

Mem Limit 524288 kB

Given an integer N , print the N -th Fibonacci number.

A Fibonacci series is a series of numbers in which each number is the sum of the two numbers preceding it. For this problem, you can assume that the first two numbers in the series are 1 and 1.

Below you can see the first few numbers of the series:

1 | 1, 1, 2, 3, 5, 8, ...

Input

The input will contain an integer N ($0 < N < 50$).

Output

Print the N -th Fibonacci number.

Example

Input	Output
3	2

Problem Y. Sets-STL

OS Linux

Sets are a part of the C++ STL. Sets are containers that store unique elements following a specific order. Here are some of the frequently used member functions of sets:

- *Declaration:*

```
1 | set<int>s; //Creates a set of integers.
```

- *Size:*

```
1 | int length=s.size(); //Gives the size of the set.
```

- *Insert:*

```
1 | s.insert(x); //Inserts an integer x into the set s.
```

- *Erasing an element:*

```
1 | s.erase(val); //Erases an integer val from the set s.
```

- *Finding an element:*

```
1 | set<int>::iterator itr=s.find(val); //Gives the iterator to the element
2 | Ex: set<int>::iterator itr=s.find(100); //If 100 is not present t
```

To know more about sets [click Here](#). Coming to the problem, you will be given Q queries. Each query is of one of the following three types:

1 x : Add an element x to the set.

2 x : Delete an element x from the set. (If the number x is not present in the set, then do nothing).

3 x : If the number x is present in the set, then print "Yes"(without quotes) else print "No"(without quotes).

Input Format

The first line of the input contains Q where Q is the number of queries. The next Q lines contain 1 query each. Each query consists of two integers y and x where y is the type of the query and x is an integer.

Constraints

$$1 \leq Q \leq 10^5$$

$$1 \leq y \leq 3$$

$$1 \leq x \leq 10^9$$

Output Format

For queries of type 3 print "Yes"(without quotes) if the number x is present in the set and if the number is not present, then print "No"(without quotes).

Each query of type 3 should be printed in a new line.

Input	Output
8	Yes
1 9	No
1 6	No
1 10	
1 4	
3 6	
3 14	
2 6	
3 6	

Problem X. Maps-STL

OS Linux

Maps are a part of the C++ STL. Maps are associative containers that store elements formed by a combination of a key value and a mapped value, following a specific order. The mainly used member functions of maps are:

- *Map Template:*

```
1 | std::map <key_type, data_type>
```

- *Declaration:*

```
1 | map<string,int>m; //Creates a map m where key_type is of type str
```

- *Size:*

```
1 | int length=m.size(); //Gives the size of the map.
```

- *Insert:*

```
1 | m.insert(make_pair("hello",9)); //Here the pair is inserted into
```

- *Erasing an element:*

```
1 | m.erase(val); //Erases the pair from the map where the key_type i
```

- *Finding an element:*

```
1 | map<string,int>::iterator itr=m.find(val); //Gives the iterator t
2 | Ex: map<string,int>::iterator itr=m.find("Maps"); //If Maps is no
```

- *Accessing the value stored in the key:*

```
1 | To get the value stored of the key "MAPS" we can do m["MAPS"] or
```

To know more about maps [click Here](#).

You are appointed as the assistant to a teacher in a school and she is correcting the answer sheets of the students. Each student can have multiple answer sheets. So the teacher has Q queries:

1 X Y : Add the marks Y to the student whose name is X .

2 X : Erase the marks of the students whose name is X .

3 X : Print the marks of the students whose name is X . (If X didn't get any marks print 0.)

Input Format

The first line of the input contains Q where Q is the number of queries. The next Q lines contain 1 query each. The first integer, *type* of each query is the type of the query. If query is of type 1, it consists of one string and an integer X and Y where X is the name of the student and Y is the marks of the student. If query is of type 2 or 3, it consists of a single string X where X is the name of the student.

Constraints

$$1 \leq Q \leq 10^5$$

$$1 \leq \textit{type} \leq 3$$

$$1 \leq |X| \leq 6$$

$$1 \leq Y \leq 10^3$$

Output Format

For queries of type 3 print the marks of the given student.

Input	Output
7 1 Jesse 20 1 Jess 12 1 Jess 18 3 Jess 3 Jesse 2 Jess 3 Jess	30 20 0

Problem W. Missing Number

Time Limit 1000 ms

Mem Limit 524288 kB

You are given all numbers between $1, 2, \dots, n$ except one. Your task is to find the missing number.

Input

The first input line contains an integer n .

The second line contains $n - 1$ numbers. Each number is distinct and between 1 and n (inclusive).

Output

Print the missing number.

Constraints

- $2 \leq n \leq 2 \cdot 10^5$

Example

Input	Output
5 2 3 1 5	4

Problem V. Weird Algorithm

Time Limit 1000 ms

Mem Limit 524288 kB

Consider an algorithm that takes as input a positive integer n . If n is even, the algorithm divides it by two, and if n is odd, the algorithm multiplies it by three and adds one. The algorithm repeats this, until n is one. For example, the sequence for $n = 3$ is as follows:

$$3 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$$

Your task is to simulate the execution of the algorithm for a given value of n .

Input

The only input line contains an integer n .

Output

Print a line that contains all values of n during the algorithm.

Constraints

- $1 \leq n \leq 10^6$

Example

Input	Output
3	3 10 5 16 8 4 2 1

Problem U. Repetitions

Time Limit 1000 ms

Mem Limit 524288 kB

You are given a DNA sequence: a string consisting of characters A, C, G, and T. Your task is to find the longest repetition in the sequence. This is a maximum-length substring containing only one type of character.

Input

The only input line contains a string of n characters.

Output

Print one integer: the length of the longest repetition.

Constraints

- $1 \leq n \leq 10^6$

Example

Input	Output
ATTCGGGA	3

Prince Macaw is flying here and there and searching for food. Her Princess is waiting for him at home. But he stopped when he was flying near Shahbag. He saw millions of people gathered there for only one demand- death sentence of war crimes. They were giving slogans. If you are not familiar with slogans let me tell you, slogan has two parts. One part of the slogan is yelled by the leader and the rest part is chorused from the mass people. For example, when the leader says — “tomar amar thikana” then other say “padma meghna jomuna”. Some of the slogans are listed below-

First Line	Second Line
ko te kader molla	tui rajakar tui rajakar
tumi ke ami ke	garo chakma bangali

Prince also joined with this mass people. When it was too late, Princess got worried about Prince. So she also went out. After some while, she found Prince on a tree at Shahbag yelling slogans in strong voice. Instead of getting mad, Princess asked Prince what are the slogans. So Prince told her a few slogans, first line and corresponding second line. So Princess got prepared. Given the list of the slogans and also the first lines Princess going to hear, print out the second line of the slogans.

Input

First line contains number of slogans, N ($0 < N \leq 20$) Prince is going to teach Princess. Hence follows N pairs of lines. First line of the pair contains first line of a slogan and the second line of the pair is corresponding second line of the slogan. Then another positive integer Q will be given ($Q \leq 100$). Hence follows Q lines. Each line contains first line of a slogan. You may assume that the slogans will contain lower case English alphabet and/or space. Both the lines of slogan will contain at least one alphabet. No two first lines in the Princes slogan list will be same. Each slogan will be at most 100 characters long. All of the query first line will be in Princes list.

Output

For each of the first line of the slogan, you need to print out corresponding second line.

Sample Input

```
3
ko te kader molla
tui rajakar tui rajakar
tumi ke ami ke
garo chakma bangali
jalo re jalo
agun jalo
2
jalo re jalo
ko te kader molla
```

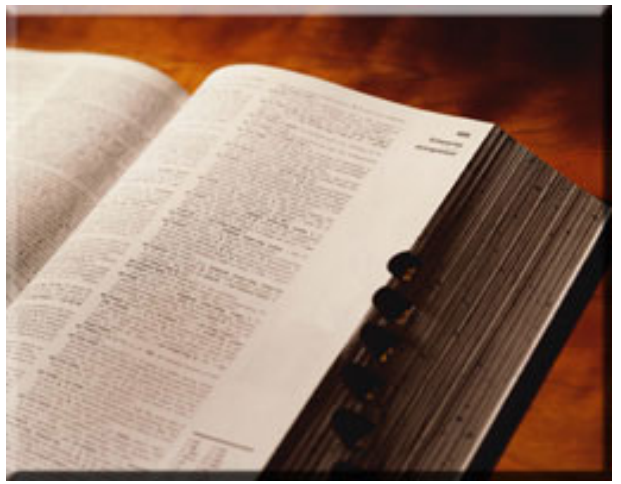
Sample Output

```
agun jalo
tui rajakar tui rajakar
```

Andy, 8, has a dream - he wants to produce his very own dictionary. This is not an easy task for him, as the number of words that he knows is, well, not quite enough. Instead of thinking up all the words himself, he has a brilliant idea. From his bookshelf he would pick one of his favourite story books, from which he would copy out all the distinct words. By arranging the words in alphabetical order, he is done! Of course, it is a really time-consuming job, and this is where a computer program is helpful.

You are asked to write a program that lists all the different words in the input text. In this problem, a word is defined as a consecutive sequence of alphabets, in upper and/or lower case.

Words with only one letter are also to be considered. Furthermore, your program must be Case Insensitive. For example, words like “Apple”, “apple” or “APPLE” must be considered the same.



Input

The input file is a text with no more than 5000 lines. An input line has at most 200 characters. Input is terminated by EOF.

Output

Your output should give a list of different words that appears in the input text, one in a line. The words should all be in lower case, sorted in alphabetical order. You can be sure that the number of distinct words in the text does not exceed 5000.

Sample Input

Adventures in Disneyland

Two blondes were going to Disneyland when they came to a fork in the road. The sign read: "Disneyland Left."

So they went home.

Sample Output

a
adventures
blondes
came
disneyland
fork
going
home
in
left
read
road
sign
so
the
they
to
two
went
were
when

Write a program that will remove all duplicates from a sequence of integers and print the list of unique integers occurring in the input sequence, along with the number of occurrences of each.

Input

The input file will contain a sequence of integers (positive, negative, and/or zero). The input file may be arbitrarily long.

Output

The output for this program will be a sequence of ordered pairs, separated by newlines. The first element of the pair must be an integer from the input file. The second element must be the number of times that that particular integer appeared in the input file. The elements in each pair are to be separated by space characters. The integers are to appear in the order in which they were contained in the input file.

Sample Input

```
3 1 2 2 1 3 5 3 3 2
```

Sample Output

```
3 4
1 2
2 3
5 1
```

Problem Q. Fair Share Settlement

Time Limit	1000 ms
Code Length Limit	50000 B
OS	Linux

You and K of your friends gather for dinner at a restaurant, and the total bill is Rs. N . You pay the entire bill, and since everyone shared the food equally, the bill needs to be split evenly among you all.

After asking your friends to reimburse their fair shares, each person pays you back. However, if the amount to be paid back wasn't an integer, they round down to the previous integer value.

For example, if each friend should have paid Rs. 13.2, they instead pay Rs. 13. Even if they should have paid Rs. 13.79, they still only pay Rs. 13, rounding down.

Calculate the net amount that you ended up paying after everyone settles their debts.

Input Format

- The first line of input will contain a single integer T , denoting the number of test cases.
- Each test case consists of one line of input, containing two space-separated integers N and K .

Output Format

For each test case, output on a new line the net amount you paid.

Constraints

- $1 \leq T \leq 10^4$
- $1 \leq N \leq 10^3$
- $1 \leq K \leq 10$

Sample 1

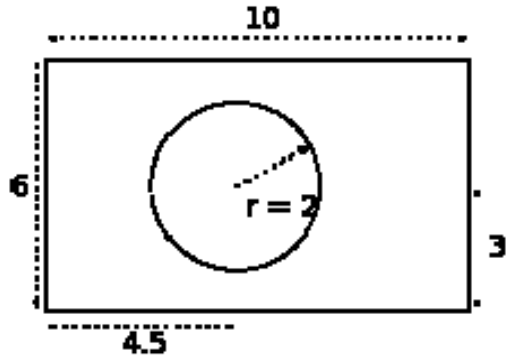
Input	Output
3 500 1 999 9 1 4	250 108 1

****Test case 1:**** You and another friend have gone for dinner, and the total cost has come to Rs. 500. You paid the entire amount of Rs. 500, and then your friend repaid you by paying back Rs. 250. Thus your net payment was $(500 - 250) = \text{Rs. } 250$.

Test case 2: Each friend's fair share was 99.9. However they round down to 99 instead. Thus your net payment was $999 - 9 \cdot 99 = \text{Rs. } 108$.

Test case 3: Each friend's fair share was 0.2, so they paid you 0. So, the entire Rs. 1 is your net payment.

The national flag of Bangladesh is bottle green in color and rectangular in size with the **length** (L) to **width** ratio of 10:6. It bears a red circle on the background of green. It maintains the **length** (L) to **radius** ratio of 5:1 (If the **length** is 10 then **width** should be 6 and **radius** should be 2). The color in the background represents the greenery of Bangladesh while the red circle symbolizes the rising sun and the sacrifice of lives in our freedom fight.



Input

First line of input will contain the number of test cases, $T \leq 100$. Then there follows T lines, each containing a positive integer $L \leq 1000$, representing length of the flag.

Output

For each test case output is a line with two space separated real numbers containing exactly two digits after decimal point. Two numbers represent the area of red and green portion respectively.

Note: Pi is considered to be $\arccos(-1)$.

Sample Input

1
10

Sample Output

12.57 47.43

Some operators check about the relationship between two values and these operators are called relational operators. Given two numerical values your job is just to find out the relationship between them that is (i) First one is greater than the second (ii) First one is less than the second or (iii) First and second one is equal.

Input

First line of the input file is an integer t ($t < 15$) which denotes how many sets of inputs are there. Each of the next t lines contain two integers a and b ($|a|, |b| < 1000000001$).

Output

For each line of input produce one line of output. This line contains any one of the relational operators '>', '<' or '=', which indicates the relation that is appropriate for the given two numbers.

Sample Input

```
3
10 20
20 10
10 10
```

Sample Output

```
<
>
=
```

Problem N. Exams

Time Limit 1000 ms

Code Length Limit 50000 B

OS Linux

In Chefland, there are X schools, and each school has Y students.
The year end results are in and a total of Z students passed the exams.

Assuming that all students appeared for the exams, find whether the number of students who passed in Chefland was **strictly greater** than 50%.

Input Format

- The first line of input will contain a single integer T , denoting the number of test cases.
- Each test case consists of three space-separated integers X , Y , and Z , as mentioned in the statement.

Output Format

For each test case, output on a new line, **YES**, if the total number of students who passed in Chefland was strictly greater than 50%, otherwise print **NO**.

You may print each character of the string in uppercase or lowercase (for example, the strings **YES**, **yEs**, **yes**, and **yeS** will all be treated as identical).

Constraints

- $1 \leq T \leq 2 \cdot 10^4$
- $1 \leq X \leq 5$
- $1 \leq Y \leq 50$
- $0 \leq Z \leq X \cdot Y$

Sample 1

Input	Output
4 2 10 12 2 10 3 1 5 3 3 6 9	YES NO YES NO

****Test case 1:**** The total number of students appeared were $2 \cdot 10 = 20$. The number of students passed were 12. Thus, number of students who passed are 60%, which is strictly greater than 50%.

Test case 2: The total number of students appeared were $2 \cdot 10 = 20$. The number of students passed were 3.

Thus, number of students who passed are 15%, which is less than 50%.

Test case 3: The total number of students appeared were $1 \cdot 5 = 5$. The number of students passed were 3.

Thus, number of students who passed are 60%, which is strictly greater than 50%.

Test case 4: The total number of students appeared were $3 \cdot 6 = 18$. The number of students passed were 9.

Thus, number of students who passed are 50%, which is equal to 50%.

Labayk Allahuma Labayk. Labayk La shareeka laka Labayk. Innal hamda wannimata laka wal mulk. La shareeka Lak

(Here I am at your service, oh Lord, here I am - here I am. No partner do you have. Here I am. Truly, the praise and the favor are yours, and the dominion. No partner do you have.)

These are the words chanted by some two million people from world heading, as if pulled by a magnet, to one single spot on Earth. As has happened every year for 14 centuries, Muslim pilgrims gather in Makkah to perform rituals based on those conducted by the Prophet Muhammad during his last visit to the city.

Performing these rituals, known as the Hajj, is the fifth pillar of Islam and the most significant manifestation of Islamic faith and unity. Undertaking the Hajj at least once is a duty for Muslims who are physically and financially able to make the journey to Makkah. The emphasis on financial ability is meant to ensure that a Muslim takes care of his family first. The requirement that a Muslim be healthy and physically capable of undertaking the pilgrimage is intended to exempt those who cannot endure the rigors of extended travel.

The pilgrimage is the religious high point of a Muslim's life and an event that every Muslim dreams of undertaking. Umrah, the lesser pilgrimage, can be undertaken at any time of the year; Hajj, however, is performed during a five-day period from the ninth through the thirteenth of Dhu Al-Hijjah, the twelfth month of the Muslim lunar calendar.

It is generally presumed that the Hajj performed on Friday is called 'Hajj-e-Akbar' and it is a superior kind of Hajj as compared with the Hajj performed on other days of the week.

But, the correct meaning of the term, as explained by a large number of the commentators of the Holy Quran is that the Umrah, which can be performed at any time throughout the year, was generally called 'Hajj-e-Asghar' (the minor Hajj). In order to distinguish hajj from Umrah the former was named 'Hajj-e-Akbar' (the greater hajj). Therefore, each and every hajj is Hajj-e-Akbar, no matter whether it is performed on Friday or on any other day. The word 'Akbar' (greater) is used only to distinguish it from Umrah which is a minor Hajj.



Input

There will be several lines in the input terminated with a line containing a single '*'. This last line should not be processed. Each of the lines will contain either Hajj or Umrah.

Output

For each line of the input, output either 'Hajj-e-Akbar' or 'Hajj-e-Asghar' in separate lines without quotations. For exact format refer to the sample.

Sample Input

```
Hajj
Umrah
Hajj
Umrah
*
```

Sample Output

```
Case 1: Hajj-e-Akbar
Case 2: Hajj-e-Asghar
Case 3: Hajj-e-Akbar
Case 4: Hajj-e-Asghar
```

Problem L. Number Line Jumps

OS Linux

You are choreographing a circus show with various animals. For one act, you are given two kangaroos on a number line ready to jump in the positive direction (i.e, toward positive infinity).

- The first kangaroo starts at location $x1$ and moves at a rate of $v1$ meters per jump.
- The second kangaroo starts at location $x2$ and moves at a rate of $v2$ meters per jump.

You have to figure out a way to get both kangaroos at the same location at the same time as part of the show. If it is possible, return **YES**, otherwise return **NO**.

Example

$$x1 = 2$$

$$v1 = 1$$

$$x2 = 1$$

$$v2 = 2$$

After one jump, they are both at $x = 3$, ($x1 + v1 = 2 + 1$, $x2 + v2 = 1 + 2$), so the answer is **YES**.

Function Description

Complete the function *kangaroo* in the editor below.

kangaroo has the following parameter(s):

- *int x1, int v1*: starting position and jump distance for kangaroo 1
- *int x2, int v2*: starting position and jump distance for kangaroo 2

Returns

- *string*: either **YES** or **NO**

Input Format

A single line of four space-separated integers denoting the respective values of $x1$, $v1$, $x2$, and $v2$.

Constraints

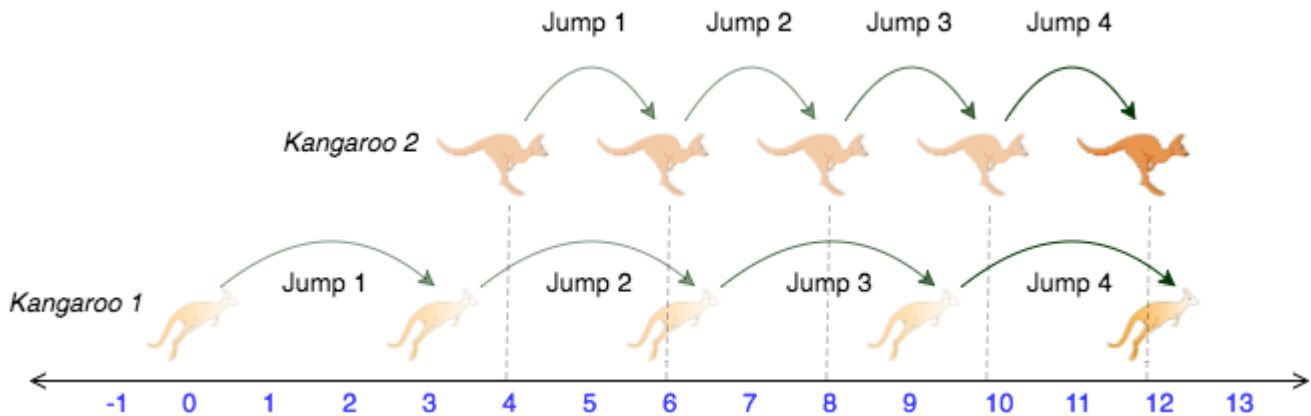
- $0 \leq x1 < x2 \leq 10000$
- $1 \leq v1 \leq 10000$

- $1 \leq v_2 \leq 10000$

Input	Output
0 3 4 2	YES

Explanation 0

The two kangaroos jump through the following sequence of locations:



From the image, it is clear that the kangaroos meet at the same location (number **12** on the number line) after same number of jumps (**4** jumps), and we print **YES**.

Input	Output
0 2 5 3	NO

Explanation 1

The second kangaroo has a starting location that is ahead (further to the right) of the first kangaroo's starting location (i.e., $x_2 > x_1$). Because the second kangaroo moves at a faster rate (meaning $v_2 > v_1$) and is already ahead of the first kangaroo, the first kangaroo will never be able to catch up. Thus, we print NO.

Problem K. Time Conversion

OS Linux

Given a time in [12-hour AM/PM format](#), convert it to military (24-hour) time.

Note: - 12:00:00AM on a 12-hour clock is 00:00:00 on a 24-hour clock.

- 12:00:00PM on a 12-hour clock is 12:00:00 on a 24-hour clock.

Example

- `s = '12:01:00PM'`

Return `'12:01:00'`.

- `s = '12:01:00AM'`

Return `'00:01:00'`.

Function Description

Complete the *timeConversion* function with the following parameter(s):

- *string s*: a time in **12** hour format

Returns

- *string*: the time in **24** hour format

Input Format

A single string *s* that represents a time in **12**-hour clock format (i.e.: **hh:mm:ssAM** or **hh:mm:ssPM**).

Constraints

- All input times are valid

Input	Output
07:05:45PM	19:05:45

Problem J. Birthday Cake Candles

OS Linux

You are in charge of the cake for a child's birthday. It will have one candle for each year of their total age. They will only be able to blow out the tallest of the candles. Your task is to count how many candles are the tallest.

Example

candles = [4, 4, 1, 3]

The tallest candles are 4 units high. There are 2 candles with this height, so the function should return 2.

Function Description

Complete the function *birthdayCakeCandles* with the following parameter(s):

- *int candles[n]*: the candle heights

Returns

- *int*: the number of candles that are tallest

Input Format

The first line contains a single integer, *n*, the size of *candles*[].

The second line contains *n* space-separated integers, where each integer *i* describes the height of *candles[i]*.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq candles[i] \leq 10^7$

Input	Output
4 3 2 1 3	2

Explanation 0

Candle heights are [3, 2, 1, 3]. The tallest candles are 3 units, and there are 2 of them.

Problem 1. Mini-Max Sum

OS Linux

Given five positive integers, find the minimum and maximum values that can be calculated by summing exactly four of the five integers. Then print the respective minimum and maximum values as a single line of two space-separated long integers.

Example

$arr = [1, 3, 5, 7, 9]$

The minimum sum is $1 + 3 + 5 + 7 = 16$ and the maximum sum is $3 + 5 + 7 + 9 = 24$. The function prints

16 24

Function Description

Complete the *miniMaxSum* function with the following parameter(s):

- $arr[5]$: an array of 5 integers

Print

Print two space-separated integers on one line: the minimum sum and the maximum sum of 4 of 5 elements. No value should be returned.

Note For some languages, like C, C++, and Java, the sums may require that you use a long integer due to their size.

Input Format

A single line of five space-separated integers.

Constraints

$$1 \leq arr[i] \leq 10^9$$

Input	Output
1 2 3 4 5	10 14

Explanation

The numbers are 1, 2, 3, 4, and 5. Calculate the following sums using four of the five integers:

1. Sum everything except **1**, the sum is $2 + 3 + 4 + 5 = 14$.
2. Sum everything except **2**, the sum is $1 + 3 + 4 + 5 = 13$.
3. Sum everything except **3**, the sum is $1 + 2 + 4 + 5 = 12$.
4. Sum everything except **4**, the sum is $1 + 2 + 3 + 5 = 11$.
5. Sum everything except **5**, the sum is $1 + 2 + 3 + 4 = 10$.

Hints: Beware of integer overflow! Use a 64-bit integer to store the sums.

Need help to get started? Try the [Solve Me First](#) problem.

Problem H. Staircase

OS Linux

Staircase detail

This is a staircase of size $n = 4$:

```
1 |  #
2 |  ##
3 |  ###
4 |  ####
```

Its base and height are both equal to n . It is drawn using `#` symbols and spaces. **The last line is not preceded by any spaces.**

Write a program that prints a staircase of size n .

Function Description

Complete the *staircase* function with the following parameter(s):

- *int n*: an integer

Print

Print a staircase as described above. No value should be returned.

Note: The last line is not preceded by spaces. All lines are right-aligned.

Input Format

A single integer, n , denoting the size of the staircase.

Constraints

$0 < n \leq 100$.

Input	Output
6	<pre> # ## ### #### ##### #####</pre>

Explanation

The staircase is right-aligned, composed of  symbols and spaces, and has a height and width of $n = 6$.

Problem G. Plus Minus

OS Linux

Given an array of integers, calculate the ratios of its elements that are *positive*, *negative*, and *zero*. Print the decimal value of each fraction on a new line with 6 places after the decimal.

Note: This challenge introduces precision problems. The test cases are scaled to six decimal places, though answers with absolute error of up to 10^{-4} are acceptable.

Example

$arr = [1, 1, 0, -1, -1]$

There are $n = 5$ elements: two positive, two negative and one zero. Their ratios are $\frac{2}{5} = 0.400000$, $\frac{2}{5} = 0.400000$ and $\frac{1}{5} = 0.200000$. Results are printed as:

```
1 | 0.400000
2 | 0.400000
3 | 0.200000
```

Function Description

Complete the *plusMinus* function with the following parameter(s):

- *int arr[n]*: an array of integers

Print

Print the ratios of positive, negative and zero values in the array. Each value should be printed on a separate line with **6** digits after the decimal. The function should not return a value.

Input Format

The first line contains an integer, n , the size of the array.

The second line contains n space-separated integers that describe $arr[n]$.

Constraints

$$0 < n \leq 100$$

$$-100 \leq arr[i] \leq 100$$

Input		Output
STDIN	Function	0.500000
-----	-----	0.333333
6	arr[] size n = 6	0.166667
-4 3 -9 0 4 1	arr = [-4, 3, -9, 0, 4, 1]	

Explanation

There are **3** positive numbers, **2** negative numbers, and **1** zero in the array.

The proportions of occurrence are positive: $\frac{3}{6} = 0.500000$, negative: $\frac{2}{6} = 0.333333$ and zeros: $\frac{1}{6} = 0.166667$.

Problem F. Diagonal Difference

OS Linux

Given a square matrix, calculate the absolute difference between the sums of its diagonals.

For example, the square matrix *arr* is shown below:

```

1 | 1 2 3
2 | 4 5 6
3 | 9 8 9

```

- The left-to-right diagonal = $1 + 5 + 9 = 15$.
- The right-to-left diagonal = $3 + 5 + 9 = 17$.

Their absolute difference is $|15 - 17| = 2$.

Function description

Complete the *diagonalDifference* function with the following parameter:

- *int arr*[*n*][*m*]: a 2-D array of integers

Return

- *int*: the absolute difference in sums along the diagonals

Input Format

The first line contains a single integer, *n*, the number of rows and columns in the square matrix *arr*.

Each of the next *n* lines describes a row, *arr*[*i*], and consists of *n* space-separated integers *arr*[*i*][*j*].

Constraints

- $-100 \leq arr[i][j] \leq 100$

Input		Output
STDIN	Function	15
-----	-----	
3	arr[][] sizes n = 3, m = 3	
11 2 4	arr = [[11, 2, 4], [4, 5, 6],	
[10, 8, -12]]		
4 5 6		
10 8 -12		

Explanation

The primary diagonal is:

$$\begin{array}{c|cc} 1 & 11 & \\ 2 & & 5 \\ 3 & & -12 \end{array}$$

Sum across the primary diagonal: $11 + 5 - 12 = 4$.

The secondary diagonal is:

$$\begin{array}{c|cc} 1 & & 4 \\ 2 & & 5 \\ 3 & 10 & \end{array}$$

Sum across the secondary diagonal: $4 + 5 + 10 = 19$

Difference: $|4 - 19| = 15$

Note: $|x|$ is the [absolute value](#) of x .

Problem E. Compare the Triplets

OS Linux

Alice and Bob each created one problem for HackerRank. A reviewer rates the two challenges, awarding points on a scale from 1 to 100 for three categories: *problem clarity*, *originality*, and *difficulty*.

The rating for Alice's challenge is the triplet $a = (a[0], a[1], a[2])$, and the rating for Bob's challenge is the triplet $b = (b[0], b[1], b[2])$.

The task is to calculate their comparison points by comparing each category:

- If $a[i] > b[i]$, then Alice is awarded 1 point.
- If $a[i] < b[i]$, then Bob is awarded 1 point.
- If $a[i] = b[i]$, then neither person receives a point.

Example

$a = [1, 2, 3]$

$b = [3, 2, 1]$

- For elements $*0*$, Bob is awarded a point because $a[0] < b[0]$.
- For the equal elements $a[1]$ and $b[1]$, no points are earned.
- Finally, for elements 2, $a[2] > b[2]$ so Alice receives a point.

The return array is $[1, 1]$ with Alice's score first and Bob's second.

Function Description

Complete the function `compareTriplets` with the following parameter(s):

- `int a[3]`: Alice's challenge rating
- `int b[3]`: Bob's challenge rating

Returns

- `int[2]`: the first element is Alice's score and the second is Bob's score

Input Format

The first line contains 3 space-separated integers, $a[0]$, $a[1]$, and $a[2]$, the respective values in triplet a .

The second line contains 3 space-separated integers, $b[0]$, $b[1]$, and $b[2]$, the respective values in triplet b .

Constraints

- $1 \leq a[i] \leq 100$
- $1 \leq b[i] \leq 100$

Input	Output
5 6 7 3 6 10	1 1

Explanation 0

In this example:

- $a = (a[0], a[1], a[2]) = (5, 6, 7)$
- $b = (b[0], b[1], b[2]) = (3, 6, 10)$

Now, let's compare each individual score:

- $a[0] > b[0]$, so Alice receives **1** point.
- $a[1] = b[1]$, so nobody receives a point.
- $a[2] < b[2]$, so Bob receives **1** point.

Alice's comparison score is **1**, and Bob's comparison score is **1**. Thus, we return the array **[1, 1]**.

Input	Output
17 28 30 99 16 8	2 1

Explanation 1

Comparing the **0th** elements, **17 < 99** so Bob receives a point.

Comparing the **1st** and **2nd** elements, **28 > 16** and **30 > 8** so Alice receives two points.

The return array is **[2, 1]**.

Problem D. Missing Number

Time Limit 1000 ms

Mem Limit 524288 kB

Given the sum of four numbers and three of those numbers, find the missing number.

All four numbers are positive integers.

Input

The input will contain one integer (the sum of the four numbers) in the first line and three integers (of those four numbers) in the second line.

All numbers in this problem fit in 32-bit signed integers.

Output

Print the missing number.

Example

Input	Output
10 1 2 3	4

Problem C. A Very Big Sum

OS Linux

Calculate and print the sum of the elements in an array.

Input Format

The first line of the input consists of an integer n .

The next line contains n space-separated integers contained in the array.

Output Format

Print the integer sum of the elements in the array.

Constraints

$$1 \leq n \leq 10$$

$$0 \leq ar[i] \leq 10^{10}$$

Sample Input

```
5
10000000001 10000000002 10000000003 10000000004 10000000005
```

Output

```
50000000015
```

Note:

The range of the 32-bit integer is **-2147483648 to 2147483647**.

When we add several integer values, the resulting sum might exceed the above range. You might need to use long long int (%lld) C/C++ to store such sums.

Problem B. Simple Array Sum

OS Linux

Given an array of integers, find the sum of its elements.

For example, if the array is $[1, 2, 3]$, then the sum is $1 + 2 + 3 = 6$, so output **6**.

Input

The first line contains the integer n denoting the length of the array.

The second line contains n space-separated integers representing the array's elements.

Output

Print a single integer: the sum of the elements of the array.

Constraints

$0 \leq n$, each element ≤ 1000

Sample Input

```
1 | 6
2 | 1 2 3 4 10 11
```

Sample Output

```
31
```

Problem A. Solve Me First

OS Linux

Read in two integers from [stdin](#), add them together, and print the answer to [stdout](#).

Input Format

There are 2 lines of input. Each line contains one integer.

Output Format

Print the sum of the integers on one line.

Note: Everything your program prints is considered “output”, so if you output some debugging statements like “Please enter an integer:”, this will be considered as part of your answer, and because it does not satisfy the output format, it will be marked wrong, even if your answer is otherwise correct!

Sample Input

```
1 | 3
2 | 4
```

Sample Output

```
7
```